

Output Data Analysis for a Single System

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Introduction

- In many simulation studies a great deal of time and money is spent on model development and “programming,” but little effort is made to analyze the simulation output data appropriately.
- As a matter of fact, a common mode of operation is to make a single simulation run of somewhat arbitrary length and then to treat the resulting simulation estimates as the “true” model characteristics.
- Since random samples from probability distributions are typically used to drive a simulation model through time, simulation estimates are just particular realizations of random variables that may have large variances.

- As a result, these estimates could, in a particular simulation run, differ greatly from the corresponding true characteristics for the model.
- The net effect is, of course, that there could be a significant probability of making erroneous inferences about the system under study.
- Historically, there are several reasons why output data analyses have not been conducted in an appropriate manner. First, some users have the unfortunate impression that simulation is largely an exercise in computer programming, albeit a complicated one.

- Consequently, some simulation “studies” begin with construction of an assumptions document and subsequent “programming,” and end with a single run of the simulation to produce the “answers.”
- In fact, however, a simulation is a computer-based statistical sampling experiment. Thus, if the results of a simulation study are to have any meaning, appropriate statistical techniques must be used to design and analyze the simulation experiments.
- A second reason for inadequate statistical analyses is that the output processes of virtually all simulations are nonstationary and autocorrelated.

- We now describe more precisely the random nature of simulation output. Let $Y_1, Y_2, \dots$ be an output stochastic process from a single simulation run.
- For example, Y_i might be the throughput (production) in the i th hour for a manufacturing system. The Y_i 's are random variables that will, in general, be neither independent nor identically distributed.

- Let $y_{11}, y_{12}, \dots, y_{1m}$ be a realization of the random variables $Y_1, Y_2, \dots, Y_m$ resulting from making a simulation run of length m observations using the random numbers $u_{11}, u_{12}, \dots$. (The i th random number used in the j th run is denoted u_{ji} .)
- If we run the simulation with a different set of random numbers $u_{21}, u_{22}, \dots$, then we will obtain a different realization $y_{21}, y_{22}, \dots, y_{2m}$ of the random variables $Y_1, Y_2, \dots, Y_m$. (The two realizations are not the same since the different random numbers used in the two runs produce different samples from the input probability distributions.)

- In general, suppose that we make n independent replications (runs) of the simulation, resulting in the observations:

$$y_{11}, \dots, y_{1i}, \dots, y_{1m}$$

$$y_{21}, \dots, y_{2i}, \dots, y_{2m}$$

$$\vdots \qquad \qquad \vdots \qquad \qquad \vdots$$

$$y_{n1}, \dots, y_{ni}, \dots, y_{nm}$$

- The observations from a particular replication (row) are clearly not IID. However, note that $y_{1i}, y_{2i}, \dots, y_{ni}$ (from the i th column) are IID observations of the random variable Y_i , for $i = 1, 2, \dots, m$. This independence across runs is the key to the relatively simple output-data-analysis methods described here.
- Then, roughly speaking, the goal of output analysis is to use the observations y_{ji} ($i = 1, 2, \dots, m; j = 1, 2, \dots, n$) to draw **inferences** about the (**distributions of the**) random variables $Y_1, Y_2, \dots, Y_m$. For example, $\bar{y}_i(n) = \sum_{j=1}^n y_{ji} / n$ is an unbiased estimate of $E(Y_i)$.

